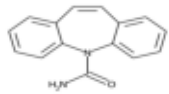


Paired Dataset

Drug structure

SMILES -> ECFP4



matched pair

Mechanism text

sodium channel
beta-2 agonist
DNA polymerase

Frozen Encoders

ECFP4 / RDKit

2048-bit fingerprint



frozen

S-Biomed-RoBERTa

CLS from each layer



frozen

Trainable Bridges

molecule bits
feature map

trainable
projection heads

W_m

2048 -> 256

text states
feature map

W_t

768 -> 256

Shared Retrieval Space

CLOOB / InfoLOOB objective

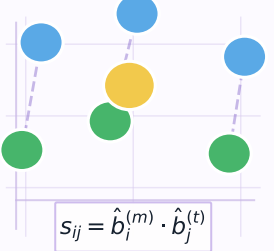
align matched pairs; repel impostors

supervises alignment

256-d joint space

cosine / dot-product retrieval

● molecule ● text ● anchor



Audit Outputs

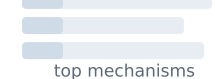
explain

ECFP4 saliency



retrieve

Ranked retrieval



probe

RoBERTa logit lens



Legend



molecule branch



text branch



frozen encoder



trainable bridge



shared space



loss objective